30 — Platform Energy & Water Integration

v1.3 — heat cascade, multi-source strategy, redundancy ladder, degraded operation

Function: Place the comfort systems (docs 10–12, 20–22) inside the platform's full energy and water architecture — vessel or land site: a high-grade process-heat source (waste-to-energy class) at the top of a heat cascade, two-stage HDH desalination as the primary water engine (where a saline/raw-water feed exists), solar thermal/electric, and a redundancy-first water ladder. **Governing reframe: redundancy outranks efficiency in the field.**

What the comfort systems actually require. Nothing in docs 10–12 or 20–22 depends on the peak grade or the identity of the upstream source. The platform's requirement is a **cascade tap at a stated grade**: 60–93 °C for the liquid track's sealed still, 60–65 °C for the solid track's regeneration (F2), and ≥ ~130 °C where a potash CO_2 bed is primary (X11). Any source that can deliver those taps continuously satisfies the integration; the prime mover is out of scope for this lineage.

---

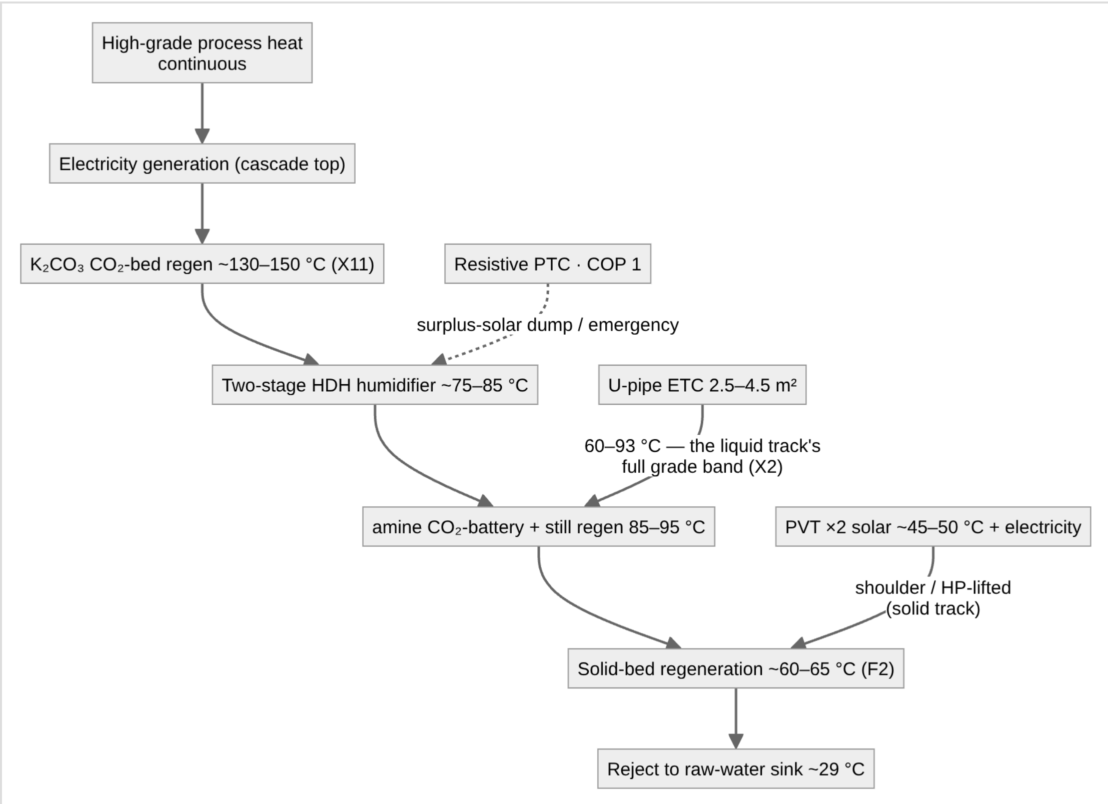
1. The two physical floors

Both stem from the ~29 °C raw-water sink (doc 00 §3):

1. **Evaporative cooling and sink-side condensation cannot beat the ~28 °C ambient dew point.** Sink-cooled atmospheric water harvesting yields ≈ 0 at DP-A; only a desiccant breaks the floor. Corollary: **the M-cycle is a cooler, not a water device.**
2. **Fresh water comes from heat (evaporate raw water → HDH/still), from air (desiccant), or from the sky (rain)** — three *physically independent* source classes; the basis of the §5 ladder.

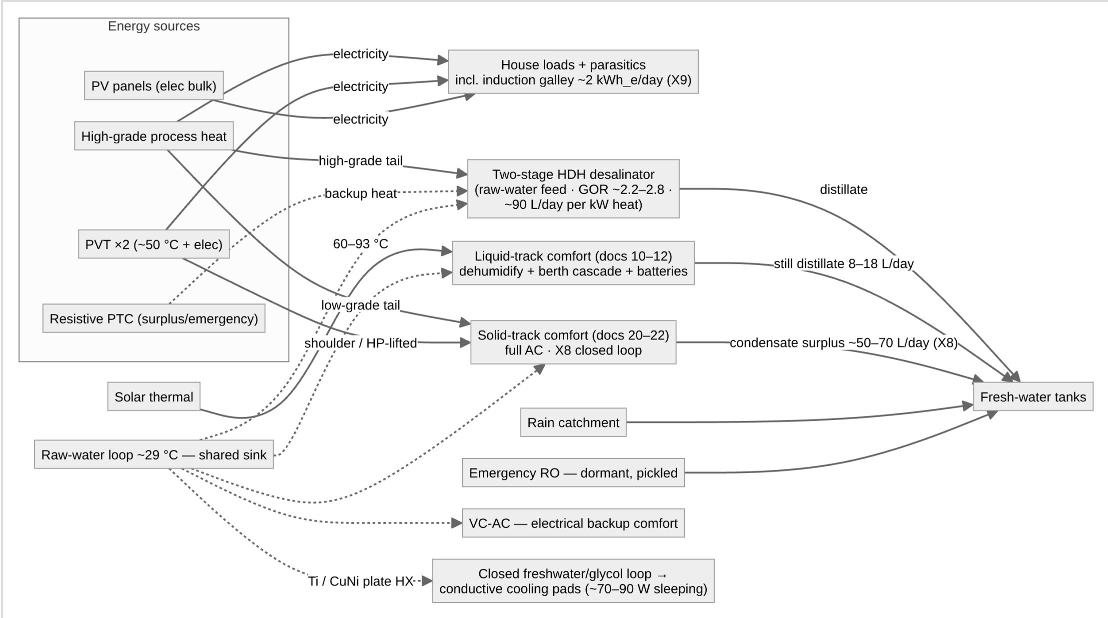
2. The heat-grade cascade (organizing principle)

Match each load to the grade of heat it needs; spend high grade only at the cascade bottom. A high-temperature source makes low-grade tail heat *free*, which is what rescues the solid track's economics (doc 20 §6) and makes the X10 CO_2 battery and X8 recovery modes essentially free to run.



Notes: PVT cannot reach HDH grade, so **solar serves comfort; the high-grade source serves water** — different subsystems. With F2, PVT-*direct* solid-bed regeneration is dead at the peak point; **the liquid track (docs 10–12) is the natural solar comfort island** because its sealed still keeps a positive driving force at any solar-grade pool temperature (X2) — this supersedes the heat-pump lift as the first fallback, with the HP lift retained behind it. Resistive heat into HDH runs ~27× RO's energy per litre — clipped-solar dump load and last resort only. The K_2CO_3 CO_2-bed rung exists only on platforms with a ≥ ~130 °C tap; at solar grade the amine bed serves (X11).

3. System architecture



Raw-water intake: marine — subsurface (5–10 m) for **stability and cleanliness, not temperature** (tropical mixed layer nearly isothermal to 20–40 m). Land — well/lake loop (often cooler: free floor depth, doc 10 §3), or an evaporative fluid cooler / dry cooler with the doc 00 §1 penalties. Surface ~29 °C water is thermodynamically sufficient for conductive body-cooling pads (skin ≈ 33–35 °C) via an isolated Ti/CuNi plate HX into a closed low-fouling loop.

HDH in one paragraph (the water engine, saline-feed sites): evaporate raw water into a closed air loop with high-grade tail heat, condense against cold raw water — membrane-free, no high pressure, feed-tolerant, hand-repairable; the dehumidifier doubles as the feed pre-heater. Realistic two-stage GOR with a 29 °C sink is ~2.2–2.8 (the warm sink pinches the cold end to ~34 °C, stranding ω ≈ 35 g/kg — lab GOR-4+ figures assume cold sinks and do not transfer). An optional desiccant polish stage recaptures the stranded fraction (+20–25%), justified only because regeneration heat is free. **HDH is never pointed at cabin air** — floor #1.

4. Sources and roles

Source	Grade	Primary role	Note
High-grade process heat	high, continuous	electricity + HDH water + regen tail	central engine; **single point of failure for water and power simultaneously**
Solar thermal (ETC 2.5–4.5 m²)	60–93 °C	**liquid-track comfort + batteries** (moisture + CO_2)	the solar comfort island, per X2
PVT ×2	~50 °C + elec	solid-track shoulder backup + parasitics	beyond two panels marginal thermal has no user (~36 kg each); optimal array = 2 PVT + rest PV
PV	elec	house/electrical bulk incl. induction galley	lighter per watt
Resistive PTC	COP 1	HDH surplus-dump + emergency	never primary on metered power

All-electric galley (X9, safety register item 2): no gas or combustion appliances in the conditioned envelope — a single burner emitted 3.6× the crew's CO_2 and ~0.25 kg/h of latent load. Induction adds ~2 kWh_e/day (trivial on a waste-heat platform; a battery/inverter sizing line on the solar-only island). Gas lockers, lines, and flame-failure devices leave the safety story entirely; the galley hood ducts into the ERV exhaust with boost-on-hood interlock.

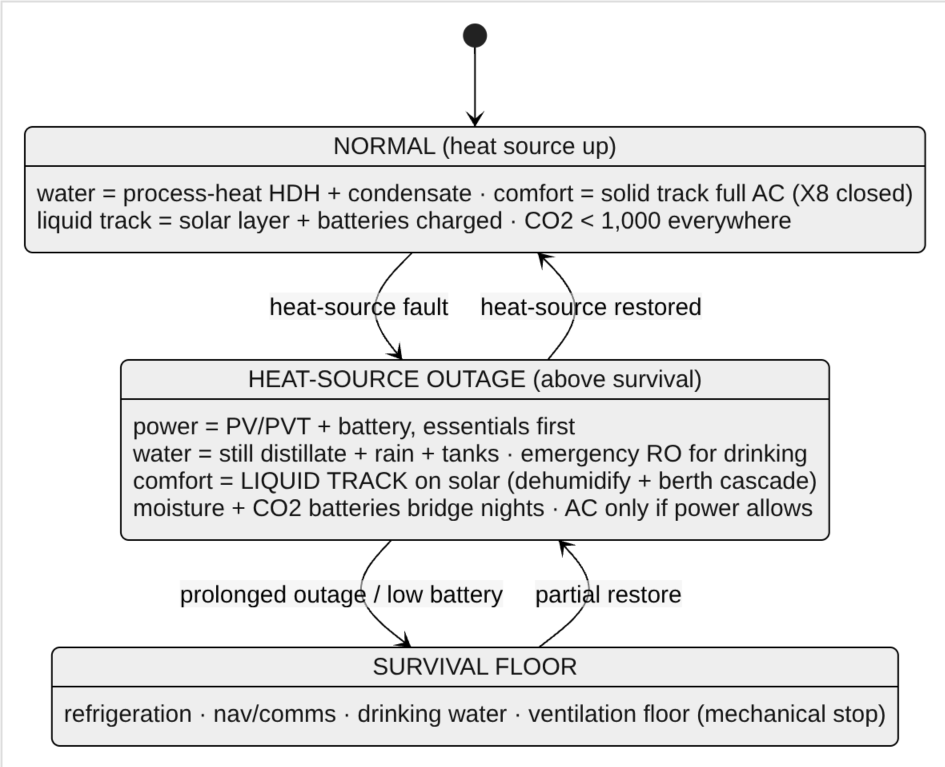
5. Water redundancy ladder

Path	Source class	Independent of HDH hardware?	Rate	Role
Process-heat HDH	raw water + free heat	— (is the HDH)	high	**primary** (saline-feed sites)
Resistive-HDH	raw water + battery elec	**No — common-mode**	high	heat-source backup only
Solid-track condensate (X8 custody)	humid air + low-grade heat	**Yes**	~50–70 L/day surplus at duty	comfort byproduct; water-neutral M-cycle in every ambient
Liquid-track still distillate	humid air + solar-grade heat	**Yes**	8–18+ L/day	the solar-independent path; also the M-cycle feed
Rain catchment	sky	**Yes**	generous, intermittent	passive; design capture into deck/roof/awning
Emergency RO (smallest unit, pickled)	raw water + small power	**Yes**	drinking + essentials	dormant; few cycles → minimal fouling; the only path robust *outside* the humid tropics
Fresh-water tanks	—	**Yes**	buffer	autonomy; bridges any outage

Common-mode warning: process-heat HDH and resistive-HDH share columns, raw-water loop, and air loop — resistive backs up *heat-source* loss only. Real security lives in the mechanically independent paths; **over-invest in the cheapest of these (tankage + rain)**. RO is retained dormant rather than primary: the maintenance objection applies to *running* RO; a pickled emergency unit inverts that calculus while preserving the one path independent of sink temperature, air humidity, weather, and the entire process-heat/HDH stack.

6. Failure philosophy & degraded operation

Goal on heat-source loss is **maintained operation above survival at a degraded level** . Essential loads: refrigeration, navigation/site comms, water production.



Recorded redundancy facts: peak-day *full AC* is waste-heat-coupled (F1/F2) — but with the liquid track as the solar layer, **dehumidified comfort and <1,000 ppm air quality survive a total heat-source outage** on 2.5–4.5 m^2 of collector, the moisture battery (~35–40 kg/night), and the CO_2 battery (solar-window regen). Where a potash bed is primary (X11), outage sealed operation requires the small amine fallback bed — the ETC's ≤93 °C cannot regenerate carbonate — otherwise degraded operation is open-mode on the ERV+DCV fallback. Manual fallback modes everywhere (hand valves, manual resistive switch, manual dampers **with the ventilation minimum stop**) so a controller fault cannot disable routing or close ventilation.

Common-mode risks: shared raw-water loop (redundant pump + passive ram-scoop/gravity feed + cross-connects + strainers); heat source (battery autonomy + independent water paths + solar comfort island); controls (manual modes, hard interlock stops); HDH mechanicals (drops both HDH paths at once — hence the independent ladder).

7. Platform-level design principles

1. Redundancy outranks efficiency in the field.
2. Match heat grade to load grade; spend high grade only at the cascade bottom.
3. Independent paths are the only real redundancy; a common-mode "backup" covers only its own specific failure.
4. Water comes from heat, air, or sky — not meaningfully from electricity.
5. Passive-first fluid movement with active understudies (vertical-axis thermosyphons, wicked heat pipes, trapped siphons, check-valved backups — rated ~30° heel for marine).
6. Over-invest in the cheap independent paths (tanks, rain).
7. Solar for the comfort layer; PV for the electrical bulk; the high-grade source for water duty.
8. Saturated exhausts end in recovery (X8); no combustion in the envelope (X9); the ventilation floor is mechanical, not software.

Version history

- **v1.0** — New lineage from archived 04: generalized to land + marine platforms; liquid track installed as the solar comfort island (supersedes HP-lift as first fallback); X8 condensate custody and X9 all-electric galley integrated; gas system removed; degraded-mode air quality and ventilation floor added to the state machine.
- **v1.1** — K_2CO_3 CO_2-bed rung (~130–150 °C, X11) added to the cascade (§2); outage disposition of the potash bed recorded (§6).
- **v1.2** — §6 state-diagram labels use the `·` separator throughout, matching the rest of the diagram and bringing the source into agreement with its `mmd_wide` layout override, which had drifted from it in punctuation. No content or claim changed; caught by `scripts/check_release.py`.
- **v1.3** — Prime-mover references genericised throughout to high-grade process heat (waste-to-energy class); the peak-temperature figure removed as unused by any SLICE derivation; the function statement gains the explicit cascade-tap interface requirement (60–93 °C liquid still · 60–65 °C solid regeneration · ≥ ~130 °C potash bed), which is what the comfort systems actually depend on. The two `mmd_wide` layout overrides were amended in step with their in-doc blocks. No figure changed.